Secure and Scalable Genomic Data Access and Analysis in Galaxy

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I. Introduction

There is an increasing need to develop a **shared framework and infrastructure** that enables the **secure access and integration of clinical human data**, in line with regulatory frameworks such as the General Data Protection Regulation (**GDPR**) and the European Health Data Space (**EHDS**), this has been showcased in ELIXIR past and current projects (**1+MG, GDI**...)

To address this, this project leverages the **EGA**, the **Galaxy** platform, and standards from the Global Alliance for Genomics and Health (**GA4GH**), particularly focusing on the use of Crypt4GH for data encryption.

II. Aim

Our aim is to **establish a secure mechanism for orchestrating access to data hosted in the EGA**, whether from its central instance or any federated nodes, **through a Galaxy instance**.

The end goal is to **ensure all output data is encrypted and only accessible to end-users via the Galaxy interface.**

Galaxy

EUROPEAN GENOME-PHENOME ARCHIVE

Global Alliance for Genomics & Health

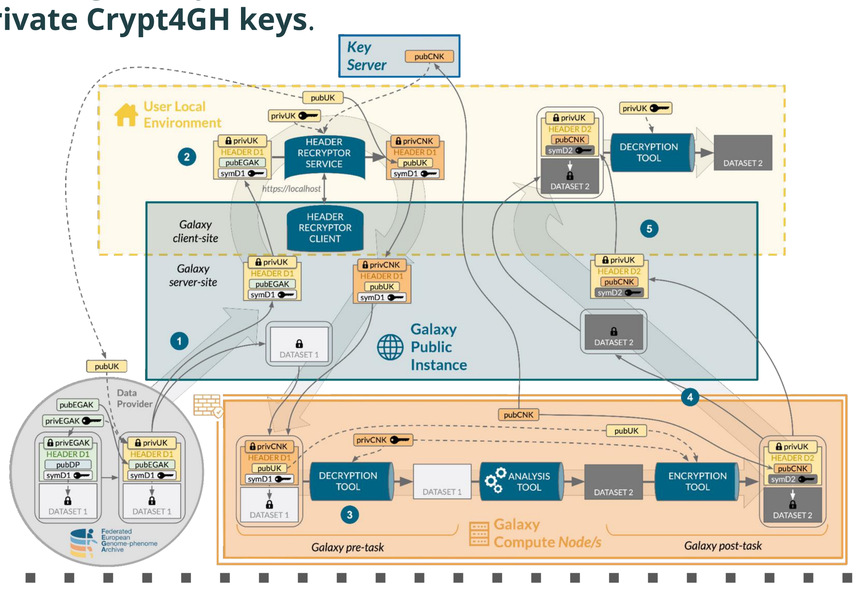
III. Objectives

- **Identify user journeys** for working with sensitive data within Galaxy
- **Enable user-orchestrated sensitive data access enable access to sensitive data from FEGA/EGA for analysis in Galaxy** based on GA4GH standards.
- Prioritise and provide for an **accessible and intuitive user experience**.
- Demonstrate complete **end-to-end solutions based on real-world use cases** from related projects (using **synthetic data**)
- Ensure that **results are communicated and available** to relevant ELIXIR platforms and users

IV. Foundation Work

2022-2023 Implementation Study: Strengthen Data Management in Galaxy

The prototype work developed within this project aimed to securely analyse (F)EGA datasets using Galaxy custom tools **without the need for the user to share their private Crypt4GH keys**.



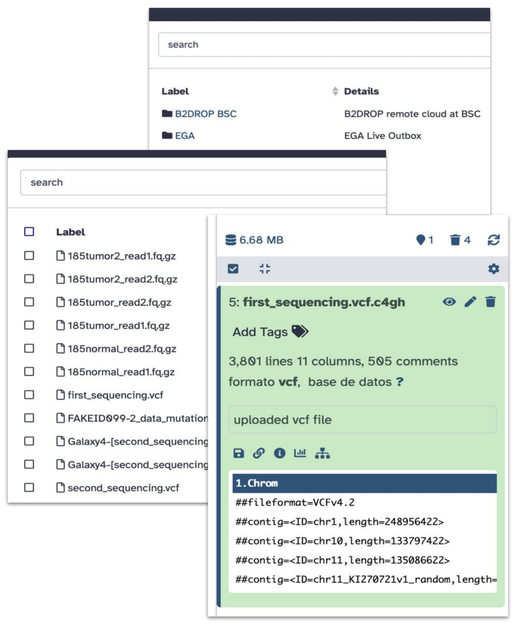
ELIXIR Europe BioHackathon 2024: Enabling Secure Data Access from Galaxy to (F)EGA

Main Goal: Allowing **secure access to sensitive data** hosted in (F)EGA from a Galaxy instance ✅

1. EGA outbox as remote core filesystem: now reachable by Galaxy ✅

2. Users are able to **see EGA encrypted datasets** they can access from Galaxy ✅

3. Those encrypted datasets can be downloaded into Galaxy and **seamlessly decrypted using the user's private Crypt4GH key** ✅



V. The Project

This work will be carried out as part of an ELIXIR Commissioned Service project within the Science tier, under the Human Data and Translational Research programme, through the project **"Empowering Users Orchestrating Sensitive Data Access for Interactive Federated Analysis in Virtual Research Environments"**

Human Data & Translational Research

VI. Organisation

Four collaborating ELIXIR Nodes: **Spain, Norway, Germany and Belgium**

TASK 1	TASK 2	TASK 3	TASK 4	TASK 5
Project management and **Coordination**, Outreach and Sustainability	Environment **Requirements and Needs Assessment**. Investigation and Adoption of Standards and Recommendations	**Crypt4GH Recryptor** Services with **Secure Management** and Trusted Exchange of Encryption Keys	Interactive **End-user Orchestration of Sensitive Data** Analysis **in Public Galaxy Servers**	**Back-end Infrastructure** for Encrypted Data Provision and **Federated Analysis in Trusted Research Environments** (TREs)

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